Week 3 at a glance

We will be learning and practicing to:

- Model systems with tools from discrete mathematics and reason about implications of modelling choices. Explore applications in CS through multiple perspectives, including software, hardware, and theory.
  - Mon – Determining the properties of positional number representations, including overflow and bit operations
  - Mon+ Wed – Connecting logical circuits and compound proposition and tracing to calcluate output values
- Translate between different representations to illustrate a concept.
  - Fri – Translating between symbolic and English versions of statements using precise mathematical language
- Use precise language to clearly present logical arguments and communicate with technical and non-technical colleagues.
  - Fri – Distinguishing between tautologies, contradictions, and contingencies
- Use precise notation to encode meaning and present arguments concisely and clearly
  - Mon+ Wed – Listing the truth tables of atomic boolean functions (and, or, xor, not, if, iff)
- Know, select and apply appropriate computing knowledge and problem-solving techniques. Reason about computation and systems. Use mathematical techniques to solve problems. Determine appropriate conceptual tools to apply to new situations. Know when tools do not apply and try different approaches. Critically analyze and evaluate candidate solutions.
  - Wed – Evaluating compound propositions
  - Fri – Judging logical equivalence of compound propositions using symbolic manipulation with known equivalences, including DeMorgan's Law
  - Fri – Judging logical equivalence of compound propositions using truth tables
  - Wed – Rewriting compound propositions using normal forms
  - Fri – Judging whether a collection of propositions is consistent

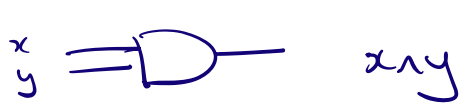
TODO:

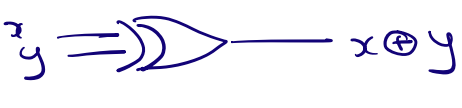
Review quiz based on Week 2 class material (due Monday)

Schedule your Test 1 and Test 2 times at PrairieTest (http://us.prairietest.com)

Review quiz based on Week 3 class material (due Monday next week)

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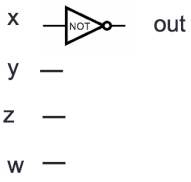




Week 3 Monday: Fixed-width Addition and Circuits

Example digital circuit:

Last time we considered this circuit:



Interpret x,y,z,w as representing $[x\,y\,z\,w]_{s,4}$

When out is 1 what can we conclude?

We know x is 0 in this case so # representing is $[0\,y\,z\,w]_{s,4} \geq 0$.

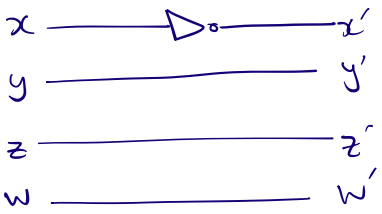
We noticed that the output of this circuit just depends on the value of x.

Design a digital circuit that, given a sign-magnitude width 4 representation of an integer, outputs a sign-magnitude width 4 represention of -1 times that integer. *Are there any corner cases where this circuit doesn't work?*

Input represents $n = [x\,y\,z\,w]_{s,4}$ (sign)

Output represents $n' = [x\,y'z'w']_{s,4}$ (sign)

Goal: $n' = -n$



Test: $n=5$, $n=1$, $n=0$

Design a digital circuit that, given a 2s complement width 4 representation of an integer, outputs a 2s complement width 4 representation of an -1 times that integer. *Are there any corner cases where this circuit doesn't work?*

Does previous circuit?

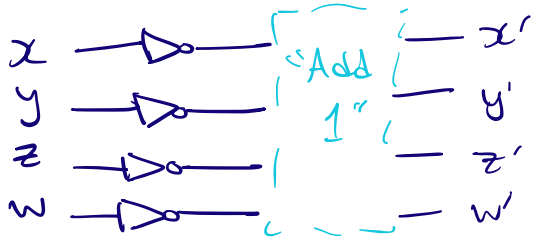
$5 = [0\,1\,0\,1]_{2c,4}$ (−8, 4, 2, 1)

Try $n=5$

$-5 = [1\,0\,1\,1]_{2c,4}$ (−8, 4, 2, 1)

Trick: "Flip bits and add 1"

In this example: 0101 → (Flip bits) 1010 → (add 1) 1011



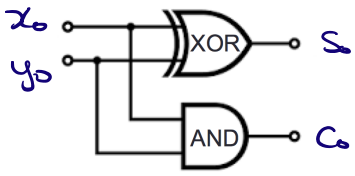
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Fixed-width addition: adding one bit at time, using the usual column-by-column and carry arithmetic, and dropping the carry from the leftmost column so the result is the same width as the summands. In many cases, this gives representation of the correct value for the sum when we interpret the summands in fixed-width binary or in 2s complement.

For single column:

Input		Output	
x_0	y_0	c_0	s_0
1	1	1	0
1	0	0	1
0	1	0	1
0	0	0	0

x_0 + y_0 = 0 (with carry!)



Addition:

Draw a logic circuit that implements binary addition of two numbers that are each represented in fixed-width binary:

- Inputs x_0, y_0, x_1, y_1 represent $(x_1x_0)_{2,2}$ and $(y_1y_0)_{2,2}$
- Outputs z_0, z_1, z_2 represent $(z_2z_1z_0)_{2,3} = (x_1x_0)_{2,2} + (y_1y_0)_{2,2}$ (may require up to width 3)

First approach: half-adder for each column, then combine carry from right column with sum of left column

Write expressions for the circuit output values in terms of input values:

$z_0 =$ $x_0 \oplus y_0$

$z_1 =$ $(x_1 \oplus y_1) \oplus (x_0 \wedge y_0)$

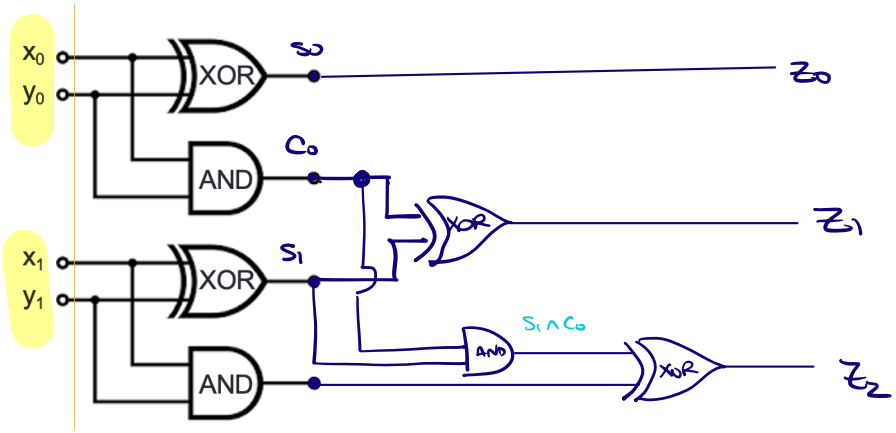
$z_2 =$ $(x_1 \wedge y_1) \oplus ((x_1 \oplus y_1) \wedge (x_0 \wedge y_0))$ (s_1, c_0)

c_0

$(x_1x_0)_{2,2}$

+ $(y_1y_0)_{2,2}$

$(z_2z_1z_0)_{2,3}$



x_1	x_0	y_1	y_0	z_2	z_1	z_0
1	1	1	1	1	1	0
1	1	1	0	1	0	1
1	1	0	1	1	0	0
⋮				⋮		
0	0	0	0	0	0	0

There are other approaches, for example: for middle column, first add carry from right column to x_1, then add result to y_1

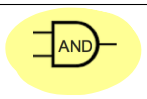
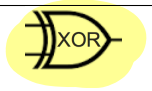
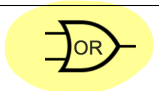
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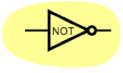
Week 3 Wednesday: Propositional Logic

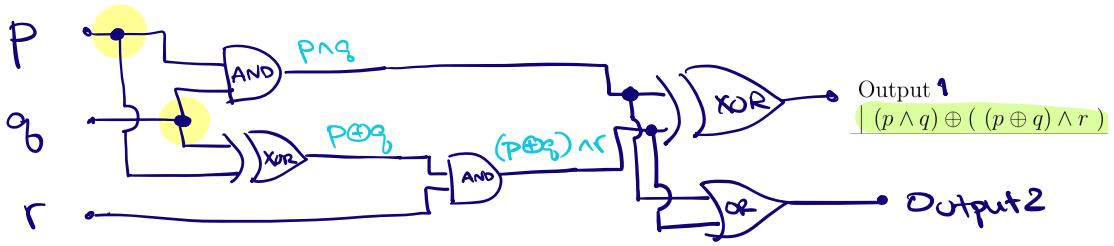
Logical operators aka propositional connectives

Conjunction	AND	$\wedge$	`\land`	2 inputs	Evaluates to T exactly when **both** inputs are T
Exclusive or	XOR	$\oplus$	`\oplus`	2 inputs	Evaluates to T exactly when **exactly one** of inputs is T
Disjunction	OR	$\vee$	`\lor`	2 inputs	Evaluates to T exactly when **at least one** of inputs is T
Negation	NOT	$\neg$	`\lnot`	1 input	Evaluates to T exactly when its input is F

Truth tables: Input-output tables where we use T for 1 and F for 0. Definition tables

Input		Output		
		Conjunction	**Exclusive or**	**Disjunction**
p	q	$p \wedge q$	$p \oplus q$	$p \vee q$
T	T	T	F	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	F
				

Input	Output
	Negation
p	$\neg p$
T	F
F	T
	



Input					Output 1	Output 2
p	q	r	$p \wedge q$	$p \oplus q$	$(p \wedge q) \oplus ((p \oplus q) \wedge r)$	$(p \wedge q) \vee ((p \oplus q) \wedge r)$
T	T	T	T	F	T	
T	T	F	T	F	T	
T	F	T	F	T	T	
T	F	F	F	T	F	
F	T	T	F	T	T	
F	T	F	F	T	F	
F	F	T	F	F	F	
F	F	F	F	F	F	

$(T \wedge T) \oplus ((T \oplus T) \wedge T)$ (T, F, T)

$F \oplus (T \wedge T)$ (F, T)

$F \oplus (T \wedge F)$ (F)

Bonus: connect this to the half adder circuit from Monday

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Given a truth table, how do we find an expression using the input variables and logical operators that has the output values specified in this table?

Application: design a circuit given a desired input-output relationship.

Input		Output	
p	q	$mystery_1$	$mystery_2$
T	T	T	F
T	F	T	F
F	T	F	F
F	F	T	T

$p \wedge q$
$p \wedge \neg q$
$\neg p \wedge q$
$\neg p \wedge \neg q$

3 rows evaluate to T

Avoid row: $\neg p \vee \neg q$
Avoid row: $\neg p \vee q$
Avoid row: $p \vee \neg q$
Avoid row: $p \vee q$

Expressions that have output $mystery_1$ are

"To avoid that p is F and q is T, could make p be T or q be F"

$$p \vee \neg q$$

"To evaluate to T, I should be in the 1st row OR in the 2nd row OR in the 3rd row"

$$(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$$

Expressions that have output $mystery_2$ are

$$(\neg p \vee \neg q) \wedge (\neg p \vee q) \wedge (p \vee \neg q)$$

$$\neg p \wedge \neg q$$

Idea: To develop an algorithm for translating truth tables to expressions, define a convenient **normal form** for expressions.

Definition An expression built of variables and logical operators is in **disjunctive normal form** (DNF) means that it is an OR of ANDs of variables and their negations.

Definition An expression built of variables and logical operators is in **conjunctive normal form** (CNF) means that it is an AND of ORs of variables and their negations.

Summary: To build the DNF, identify rows with output T and write an expression that captures truth-value settings that land in at least one of these rows $(\text{row } i_1) \vee (\text{row } i_2) \vee \cdots \vee (\text{row } i_N)$

To build the CNF, identify rows with output F and write an expression that captures truth-value settings that avoid all of these rows $(\neg \text{row } j_1) \wedge (\neg \text{row } j_2) \wedge \cdots \wedge (\neg \text{row } j_M)$

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Week 3 Friday: Logical Equivalence

Proposition: Declarative sentence that is true or false (not both).

Propositional variable: Variable that represents a proposition.

Compound proposition: New proposition formed from existing propositions (potentially) using logical operators. *Note*: A propositional variable is one example of a compound proposition.

Truth table: Table with one row for each of the possible combinations of truth values of the input and an additional column that shows the truth value of the result of the operation corresponding to a particular row.

Logical equivalence : Two compound propositions are **logically equivalent** means that they have the same truth values for all settings of truth values to their propositional variables.

Tautology: A compound proposition that evaluates to true for all settings of truth values to its propositional variables; it is abbreviated T.

Contradiction: A compound proposition that evaluates to false for all settings of truth values to its propositional variables; it is abbreviated F.

Contingency: A compound proposition that is neither a tautology nor a contradiction.

Label each of the following as a tautology, contradiction, or contingency.

$p \wedge p$

$p \oplus p$

$p \vee p$

$p \vee \neg p$

$p \wedge \neg p$

Extra Example: Which of the compound propositions in the table below are logically equivalent?

Input		Output				
p	q	$\neg(p \wedge \neg q)$	$\neg(\neg p \vee \neg q)$	$(\neg p \vee q)$	$(\neg q \vee \neg p)$	$(p \wedge q)$
T	T					
T	F					
F	T					
F	F					

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(Some) logical equivalences

Can replace p and q with any compound proposition

$\neg(\neg p) \equiv p$				**Double negation**
$p \vee q \equiv q \vee p$		$p \wedge q \equiv q \wedge p$		**Commutativity** Ordering of terms
$(p \vee q) \vee r \equiv p \vee (q \vee r)$		$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$		**Associativity** Grouping of terms
$p \wedge F \equiv F$	$p \vee T \equiv T$	$p \wedge T \equiv p$	$p \vee F \equiv p$	**Domination** aka short circuit evaluation
$\neg(p \wedge q) \equiv \neg p \vee \neg q$		$\neg(p \vee q) \equiv \neg p \wedge \neg q$		**DeMorgan's Laws**

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